*Chemistry 3A*

Introductory General Chemistry

# *Experiment 7a*

# Calorimetry

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# Introduction

- **Calorimetry** means the measurement of heat. This measurement is of heat energy either given off (exothermic) or absorbed (endothermic) by the system from the surroundings
- Heat is measured by **temperature** changes with a thermometer and utilizes the specific heat capacity equation:

$$q = m \times c \times \Delta T$$

| Symbol | Meaning | Units |
|---|---|---|
| ***q*** | energy as/in the form of heat | J or cal |
| ***m*** | mass of the substance in the system | g |
| ***c*** | specific heat capacity of substance: this is a physical property determined by scientists doing experiments | J / (g °C) |
| ***ΔT*** | change in temperature in °C: this is the final temperature minus the initial temperature | °C |

# Background: Part 1

When a solid is dissolved in a liquid (solvent), there can be an energy change related to the molecules of the solid interacting with the molecules of the solvent.

- When heat is released in dissolving, reaction is exothermic
- When heat is absorbed in dissolving, reaction is endothermic

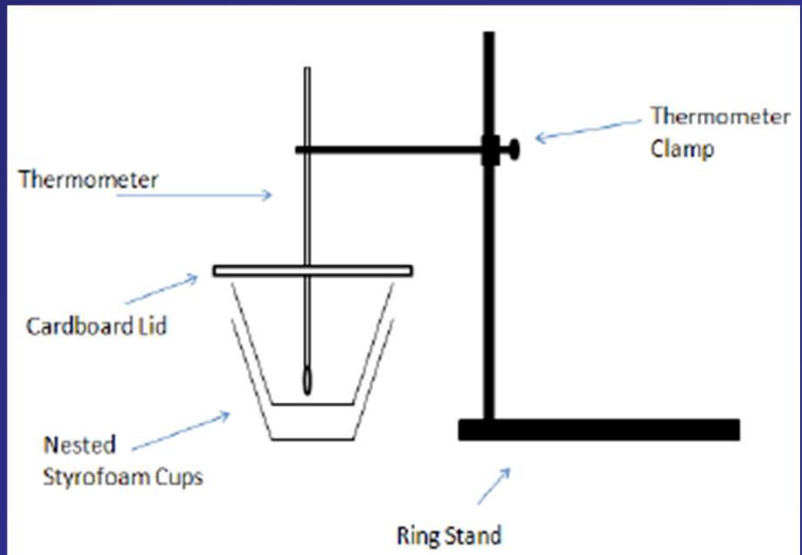


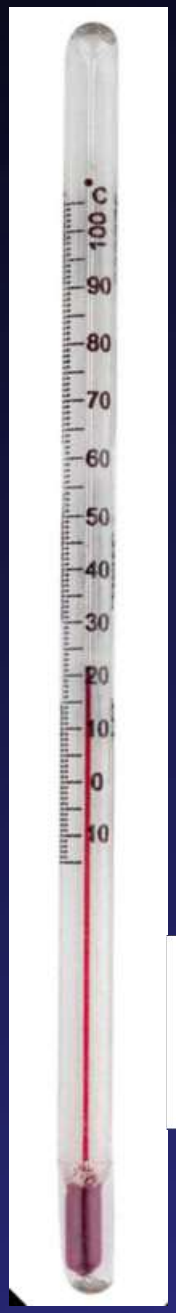

**Figure 1: Calorimeter Set-**

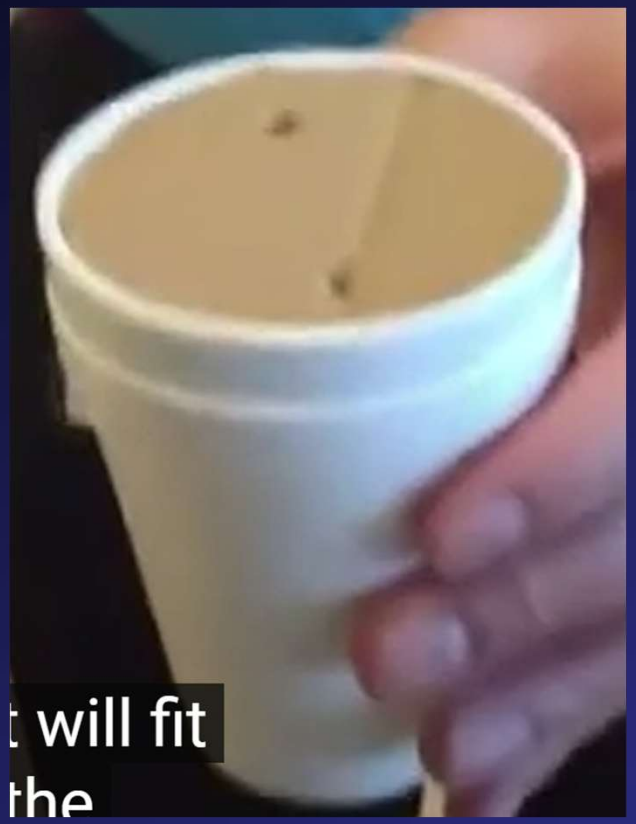
# Background: Part 2

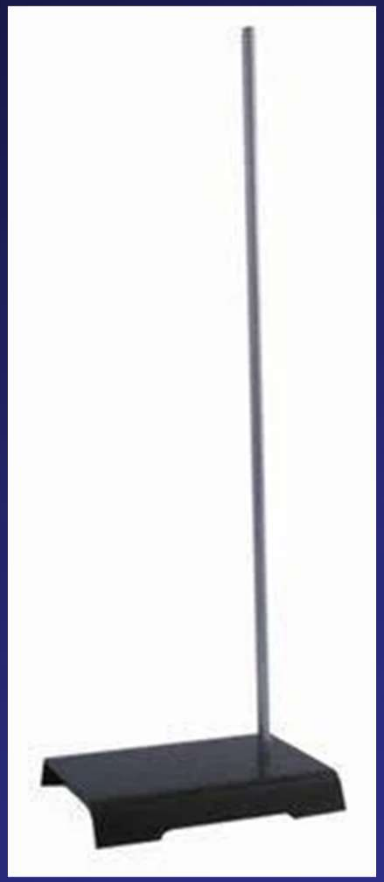
Objects with different temperatures in contact with each other, heat from the warmer object transfers to the cooler object until both reach same temperature.

- Since energy is conserved, if $H_2O$ as ice is put in contact with $H_2O$ as water, it should be possible to measure temperature changes in the water as ice melts to see if this law is observed

# Equipment You Will Use










# Consumables



# Procedure: Part 1

1. Construct the calorimetry setup with the foam cups
2. Measure the mass of the cups ONLY
3. Add 45.0 to 55.0 mL DI water to grad cylinder & pour into cups
4. Record temperature to nearest 0.1°C
5. On glassine paper, scoop out 4.8-5.2 g citric acid without causing excessive waste. Tighten the reagent lid after.
6. Add citric acid carefully but quickly to the calorimeter with water and place lid on setup; VERY CAREFULLY with lid closed stir the solution until solid dissolves

# Procedure: Part 1

7. Record temperature to nearest 0.1°C after it stabilizes (it might go up or down)
8. Remove the cardboard lid. Record the mass of the calorimeter setup: cups + solution
9. Empty the solution into a beaker (for waste) then rinse and dry the inner cup
10. Switch the inner cup with outer cup
11. Repeat steps 3-8 with sodium carbonate ($Na_2CO_3$)
12. Pour out the waste into the beaker, discard the solution into the sink

# Procedure: Part 2

1. Construct the calorimetry setup with the foam cups and add 45.0-55.0 mL DI water
2. Measure the mass of calorimeter with water without lid
3. Record temperature to nearest 0.1°C
4. Transfer 2-3 cubes ice to calorimeter
5. Place lid and stir carefully with thermometer until ice melts. Record the final temperature
6. Record mass of calorimeter with contents (not lid)

# Clean Up

- Return equipment cleaned if necessary to its storage areas
- Return cardboard lid to supply bin
- Discard the Styrofoam cups

# Example Data Analysis

## PART 1 – Exothermic and Endothermic Changes

**DATA**

Mass of calorimeter (empty): 16.4831 g

| | $H_3C_6H_5O_7$ | $Na_2CO_3$ |
|---|---|---|
| Initial Temperature (°C) | 22.8°C | 22.1°C |
| Final Temperature (°C) | 19.3°C | 27.9°C |
| Mass of calorimeter *and* solution (g) | 72.0944 g | 68.3891 g |

## PART 2 – Specific Heat Capacity and Enthalpy of Fusion

**DATA**

| | Data |
|---|---|
| Mass of calorimeter + water (g) | 64.3499 g |
| Initial Temperature of water (°C) | 21.4°C |
| Final Temperature of water (°C) | 15.8°C |
| Mass of calorimeter + water + ice (g) | 67.8851 g |

# Example Data Analysis

## PART 1 – Exothermic and Endothermic Changes

**DATA**

Mass of calorimeter (empty): 16.4831 g

| | $H_3C_6H_5O_7$ | $Na_2CO_3$ |
|---|---|---|
| Initial Temperature (°C) | 22.8°C | 22.1°C |
| Final Temperature (°C) | 19.3°C | 27.9°C |
| Mass of calorimeter *and* solution (g) | 72.0944 g | 68.3891 g |

**CALCULATIONS – PART 1 – EXOTHERMIC AND ENDOTHERMIC CHANGES**

* Show your work, complete with units for $H_3C_6H_5O_7$. I will assume you did $Na_2CO_3$ the same way

1. ☐ Mass of the solution

**How do you calculate?**

2. ☐ Change in Temperature of Solution

3. ☐ Heat of Solution ($q_{solution}$)

**RESULTS – PART 1 – ENDOTHERMIC AND EXOTHERMIC CHANGES**

| | $H_3C_6H_5O_7$ | $Na_2CO_3$ |
|---|---|---|
| Mass of Solution (m) **How do you calculate?** | | |
| Specific Heat of Solution (c) | 4.184 J/g°C | 4.184 J/g°C |
| Change in Temperature of Solution($\Delta T$) | | |
| Heat of solution ($q_{solution}$) | | |
| Heat of reaction* ($q_{reaction} + q_{solution} = 0$) | | |
| Reaction is: Exothermic or Endothermic | | |

* The water in the calorimeter is considered the reaction's surroundings, therefore if heat is released by the reaction, it is absorbed by the water and vice-versa.

# Example Data Analysis

## PART 2 – Specific Heat Capacity and Enthalpy of Fusion

**DATA**

| | Data |
|---|---|
| Mass of calorimeter + water (g) | 64.3499 g |
| Initial Temperature of water (°C) | 21.4°C |
| Final Temperature of water (°C) | 15.8°C |
| Mass of calorimeter + water + ice (g) | 67.8851 g |

**CALCULATIONS – PART 2 – SPECIFIC HEAT CAPACITY AND ENTHALPY OF FUSION**

* Show your work, complete with units.

1. Mass of $H_2O$

**How do you calculate?**

2. Change in Temperature of Water

3. $q_{water}$

**How do you calculate?**

4. Mass of Ice

5. Moles of Ice

6. $q_{ice}$

7. $q_{water} + q_{ice}$

**RESULTS**

| | Results |
|---|---|
| Mass of $H_2O$ | |
| Specific Heat of water (c) | 4.184 J/g°C |
| Change in Temperature of water ($\Delta T$) ($\Delta T = T_f - T_i$) | |
| $q_{water}$ (kJ) | |
| Mass of Ice | |
| Moles of Ice (n) | |
| Enthalpy of Fusion ($\Delta H_{fus}$) | 6.01 kJ/mol |
| $q_{ice}$ (kJ) | |
| $q_{water} + q_{ice}$ | |

**How do you calculate?**

# Example Data Analysis

## PART 1 – Exothermic and Endothermic Changes

**DATA**

Mass of calorimeter (empty): 16.4831 g

| | $H_3C_6H_5O_7$ | $Na_2CO_3$ |
|---|---|---|
| Initial Temperature (°C) | 22.8°C | 22.1°C |
| Final Temperature (°C) | 19.3°C | 27.9°C |
| Mass of calorimeter *and* solution (g) | 72.0944 g | 68.3891 g |

**CALCULATIONS – PART 1 – EXOTHERMIC AND ENDOTHERMIC CHANGES**

* Show your work, complete with units for $H_3C_6H_5O_7$. I will assume you did $Na_2CO_3$ the same way

1. ☐ Mass of the solution 72.0944-16.4381 = 55.6563 g

2. ☐ Change in Temperature of Solution 19.3 - 22.8 = -3.5°C

3. ☐ Heat of Solution ($q_{solution}$)

4.184 J/g°C x 55.6553 g x (-3.5°C) = -820 J 2 significant digits

**RESULTS – PART 1 – ENDOTHERMIC AND EXOTHERMIC CHANGES**

| | $H_3C_6H_5O_7$ | $Na_2CO_3$ |
|---|---|---|
| Mass of Solution (m) | 55.6553 g | 51.9060 g |
| Specific Heat of Solution (c) | 4.184 J/g°C | 4.184 J/g°C |
| Change in Temperature of Solution($\Delta T$) | -3.5°C | 5.8°C |
| Heat of solution ($q_{solution}$) | -820 J | 1300 J |
| Heat of reaction* ($q_{reaction} + q_{solution} = 0$) | 820 J | -1300 J |
| Reaction is: Exothermic or Endothermic | endothermic | exothermic |

* The water in the calorimeter is considered the reaction's surroundings, therefore if heat is released by the reaction, it is absorbed by the water and vice-versa.

# Example Data Analysis

## PART 2 – Specific Heat Capacity and Enthalpy of Fusion

**DATA**

| | Data |
|---|---|
| Mass of calorimeter + water (g) | 64.3499 g |
| Initial Temperature of water (°C) | 21.4°C |
| Final Temperature of water (°C) | 15.8°C |
| Mass of calorimeter + water + ice (g) | 67.8851 g |

**CALCULATIONS – PART 2 – SPECIFIC HEAT CAPACITY AND ENTHALPY OF FUSION**

* Show your work, complete with units.

1. Mass of $H_2O$

   64.3499 - 16.4831 = 47.8668 g

2. Change in Temperature of Water

   15.8 - 21.4 = -5.6°C

3. $q_{water}$

   47.8668 g × (-5.6°C) × 4.184 J/g°C
   = -1.1 kJ
   =1.1215 kJ (no intermediate round off)

4. Mass of Ice

   67.8851 - 64.3499 = 3.5352 g

5. Moles of Ice

   3.5352 g / 18.02 g/mol
   = 0.1962 mol

6. $q_{ice}$

   0.1962 mol × 6.01 kJ/mol
   = 1.179 kJ

7. $q_{water} + q_{ice}$

   -1100 + 1179 = 79 J
   -1121.5 + 1179 = 57 J

# Example Data Analysis

**PART 2 – Specific Heat Capacity and Enthalpy of Fusion**

**DATA**

| | Data |
|---|---|
| Mass of calorimeter + water (g) | 64.3499 g |
| Initial Temperature of water (°C) | 21.4°C |
| Final Temperature of water (°C) | 15.8°C |
| Mass of calorimeter + water + ice (g) | 67.8851 g |

**RESULTS**

| | Results |
|---|---|
| Mass of $H_2O$ | 47.8668 g |
| Specific Heat of water (c) | 4.184 J/g°C |
| Change in Temperature of water ($\Delta T$) ($\Delta T = T_f - T_i$) | -5.6°C |
| $q_{water}$ (kJ) | -1.122 kJ |
| Mass of Ice | 3.5352 g |
| Moles of Ice (n) | 0.1962 mol |
| Enthalpy of Fusion ($\Delta H_{fus}$) | 6.01 kJ/mol |
| $q_{ice}$ (kJ) | 1.179 kJ |
| $q_{water} + q_{ice}$ | 57 J |